3.8 PRACTICE SET

3.8.1 QUIZZES

A set of interactive quizzes for this chapter can be found on the book's website. It is strongly recommended that the student takes the quizzes to check his/her understanding of the materials before continuing with the practice set.

3.8.2 REVIEW QUESTIONS

Q3-1. Name five types of data that a computer can process.

Q3-2. How is bit pattern length related to the number of symbols the bit pattern can represent?

Q3-3. How does the bitmap graphic method represent an image as a bit pattern?

Q3-4. What is the advantage of the vector graphic method over the bitmap graphic method? What is the disadvantage?

Q3-5. What steps are needed to convert audio data to bit patterns?

Q3-6. Compare and contrast the representation of positive integers in unsigned, sign-and-magnitude format, and two's complement format.

Q3-7. Compare and contrast the representation of negative integers in sign-and-magnitude and two's complement format.

Q3-8. Compare and contrast the representation of zero in sign-and-magnitude, two's complement, and Excess formats.

Q3-9. Discuss the role of the leftmost bit in sign-and-magnitude, and two's complement formats.

Q3-10. Answer the following questions about floating-point representations of real numbers:

- **a.** Why is normalization necessary?
- **b.** What is the mantissa?
- **c.** After a number is normalized, what kind of information does a computer store in memory?

3.8.3 PROBLEMS

P3-1. How many distinct 5-bit patterns can we have?

P3-2. In some countries vehicle license plates have two decimal digits (0 to 9). How many distinct plates can we have? If the digit 0 is not allowed on the license plate, how many distinct plates can we have?

P3-3. Redo Problem P3-2 for a license plate that has two digits followed by three uppercase letters (A to Z).

P3-4. A machine has eight different cycles. How many bits are needed to represent each cycle?

P3-5. A student's grade in a course can be A, B, C, D, F, W (withdraw), or I (incomplete). How many bits are needed to represent the grade?

P3-6. A company has decided to assign a unique bit pattern to each employee. If the company has 900 employees, what is the minimum number of bits needed to create this system of representation? How many patterns are unassigned? If the company hires another 300 employees, should it increase the number of bits? Explain your answer.

P3-7. If we use a 4-bit pattern to represent the digits 0 to 9, how many bit patterns are wasted?

P3-8. An audio signal is sampled 8000 times per second. Each sample is represented by 256 different levels. How many bits per second are needed to represent this signal?

P3-9. Change the following decimal numbers to 8-bit unsigned integers.

- **a.** 23
- **b.** 121
- **c.** 34
- **d.** 342

P3-10. Change the following decimal numbers to 16-bit unsigned integers.

a. 41
b. 411
c. 1234
d. 342

P3-11. Change the following decimal numbers to 8-bit two's complement integers.

a. -12
b. -145
c. 56
d. 142

P3-12. Change the following decimal numbers to 16-bit two's complement integers.

a. 102
b. -179
c. 534
d. 62,056

P3-13. Change the following 8-bit unsigned numbers to decimal.

a. 01101011
b. 10010100
c. 00000110
d. 01010000

P3-14. Change the following 8-bit two's complement numbers to decimal.

a. 01110111
b. 11111100
c. 01110100
d. 11001110

P3-15. The following are two's complement binary numbers. Show how to change the sign of the number.

a. 01110111
b. 11111100
c. 01110111
d. 11001110

P3-16. If we apply the two's complement operation to a number twice, we should get the original number. Apply the two's complement operation to each of the following numbers and see if we can get the original number.

a. 01110111
b. 11111100
c. 01110100
d. 11001110

P3-17. Normalize the following binary floating-point numbers. Explicitly show the value of the exponent after normalization.

a. 1.10001
b. $2^3 \times 111.1111$
c. $2^{-2} \times 101.110011$
d. $2^{-5} \times 101101.00000110011000$

P3-18. Convert the following numbers in 32-bit IEEE format.

a. $-2^0 \times 1.10001$
b. $+2^3 \times 1.111111$
c. $+2^{-4} \times 1.01110011$
d. $-2^{-5} \times 1.01101000$

P3-19. Convert the following numbers in 64-bit IEEE format.

a. $-2^0 \times 1.10001$
b. $+2^3 \times 1.111111$
c. $+2^{-4} \times 1.01110011$
d. $-2^{-5} \times 1.01101000$

P3-20. Convert the following numbers in 32-bit IEEE format.

a. 7.1875
b. -12.640625
c. 11.40625
d. -0.375

P3-21. The following are sign-and-magnitude binary numbers in a 8-bit allocation. Convert them to decimal.

a. 01110111
b. 11111100
c. 01110100
d. 11001110

P3-22. Convert the following decimal integers to sign-and-magnitude with 8-bit allocation.

a. 53
b. -107
c. -5
d. 154

P3-23. One method of representing signed numbers in a computer is one's complement representation. In this representation, to represent a positive number, we store the binary number. To represent a negative number, we apply the one's complement operation on the number. Store the following decimal integers to one's complement with 8-bit allocation.

a. 53
b. -107
c. -5
d. 154

P3-24. The following are one's complement binary numbers in a 8-bit allocation. Convert them to decimal (see P3-23).

- **a.** 01110111
- **b.** 11111100
- **c.** 01110100
- **d.** 11001110

P3-25. If we apply the one's complement operation to a number twice, we should get the original number. Apply the one's complement operation twice to each of the following numbers and see if you can get the original number (see P3-23).

- **a.** 01110111
- **b.** 11111100
- **c.** 01110100
- **d.** 11001110

P3-26. An alternative method to find the two's complement of a number is to first take the one's complement of the number (see P3-23) and then add 1 to the result. (Adding binary integers is explained in Chapter 4). Try both methods using the following numbers. Compare and contrast the results.

- **a.** 01110111
- **b.** 11111100
- **c.** 01110100
- **d.** 11001110

P3-27. The equivalent of one's complement (see P3-23) in the binary system is nine's complement in the decimal system (1 =2-1 and 9 = 10 - 1). With n-digit allocation, we can represent nine's complement numbers in the range of: -[(10V2) - 1] to + [(10V2 - 1)]. The nine's complement of a number with *n* digit allocation is obtained as follows. If the number is positive, the nine's complement of the number is itself. If the number is negative, we subtract each digit from 9. Answer the following questions for three-digit allocation:

- **a.** What is the range of the numbers we can represent using nine's complement?
- **b.** In this system, how can we determine the sign of a number?
- **c.** Do we have two zeros in this system?
- **d.** If the answer to c. is yes, what is the representation for +0 and -0?

P3-28. Assuming three-digit allocation, find the nine's complement of the following decimal numbers (see P3-27).

- **a.** +234
- **b.** +560
- **c.** -125
- **d.** -111

P3-29. The equivalent of two's complement in the binary system is ten's complement in the decimal system (in the binary system, 2 is the base, in the decimal system, 10 is the base). Using n-digit allocation, we can represent numbers in the range of: -[10^n/2) to + [10^n/2-1) in ten's complement format. The ten's complement of a number with n-digit allocation is obtained by first finding the nine's complement of the

allocation is obtained by first finding the nine's complement of the number (as described in P3-27) and then adding 1 to the result. Answer the following questions for three-digit allocation.

a. What is the range of the numbers we can represent using ten's complement?
b. In this system, how can we determine the sign of a number?
c. Do we have two zeros in this system?
d. If the answer to c. is yes, what is the representation for +0 and -0?

P3-30. Assuming three-digit allocation, find the ten's complement of the following decimal numbers. (The information for doing this problem is presented in P3-29.)

a. +234
b. +560
c. -125
d. -111

P3-31. The equivalent of one's complement (see P3-23) in the binary system is fifteen's complement in the hexadecimal system (1=2 - 1 and 15= 16 - 1).

a. What range of numbers can we represent with three-digit allocation in fifteen's complement?
b. Explain how the fifteen's complement of a number is obtained in the hexadecimal system.
c. Do we have two zeros in this system?
d. If the answer to c. is yes, what is the representation for +0 and -0?

P3-32. Assuming three-digit allocation, find the fifteen's complement of the following hexadecimal numbers.

a. +B14
b. +FE1
c. -1A
d. -1E2

P3-33. The equivalent of two's complement in the binary system is sixteen's complement in the hexadecimal system. Read the explanation provided for P3-30 to answer the following questions:

a. What range of numbers can we represent with three-digit allocation in sixteen's complement?
b. Explain how a sixteen's complement of a number is obtained in the hexadecimal system.
c. Do we have two zeros in this system?
d. If the answer to c. is yes, what is the representation for +0 and -0?

P3-34. Assuming three-digit allocation, find the sixteen's complement of the following hexadecimal numbers (see P3-33).

a. +B14
b. +FE1
c. -1A
d. -1E2